

Ella, I did read the proposal from your student. Since this is not an area of expertise for me, but I had some hunches about its limitations, I had a conversation with Gemini (my institution version that does not use data for training purposes).

My initial instincts (the small dataset and small validated dataset) prove to be roadblocks for the proposed paper. At the end there are suggestions for how to still make a paper out of this but it will require to expand both the overall dataset and the manually labeled dataset.

The conversation below is quite interesting (though I have not verified all the advice). I wonder if he can submit a "poster / short paper", not a full paper instead.

Eni

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Prompt 1

Prompt 1: I am wondering whether the claims about novelty hold here. (the pdf of the proposal was also uploaded)

Based on a close reading of your paper outline, **yes, your claim to novelty holds up quite well**, provided you maintain the exact focus you have mapped out.

You have done an excellent job of preemptively dismantling the "weak" arguments for novelty (the data pipeline, using BERT, or running a Prophet forecast) and explicitly pointing out why they aren't enough on their own. By recognizing that these merely apply established methods to a new dataset, you save your paper from a quick rejection by rigorous reviewers.

The true core of your novelty rests on a highly specific methodological intersection. Here is an evaluation of why that claim is defensible, along with a few areas where reviewers might push back.



Why the Novelty Claim Holds

Your contribution relies on a "bricolage" of fields: bringing **psychometric measurement-equivalence tools** into **applied NLP / computational social science** to treat **platforms as survey modes**.

Your claim to uniqueness is valid because you are targeting a distinct gap in the literature:

- **Beyond Simple Correlation:** As you noted, papers like Pellert et al. (2022) stop at proving cross-platform sentiment correlates with real-world mood. They do not isolate

whether differences in sentiment are driven by the platform's "house style" or genuine disagreement.

- **Machine-Labelled Text vs. Human Ratings:** *Han & Anderson (2025)* look at platform measurement bias, but they use structured, explicit human data (star ratings). Your focus is on the messy, implicit signals of machine-classified text.
- **Platform as the Instrument:** *Sachdeva et al. (2022)* use Item Response Theory (IRT) to look at *annotator* bias, and *Cernat et al. (2025)* look at survey vs. digital trace data. No one has yet treated the individual social media platform itself as the formal measurement instrument for text sentiment.

By framing the platform as a "survey mode" (e.g., Twitter is the new "telephone survey" while Reddit is "face-to-face"), you turn a vague observation into something mathematically testable using generalizability theory and differential item functioning (DIF).

Potential Reviewer Pushbacks to Prepare For

While the combination is novel, reviewers at a top-tier conference like ICWSM are going to scrutinize the execution of this framing:

1. **The "Who Posts" (Selection Bias) Confound:** Reviewers will heavily probe how cleanly you can truly separate **how it is measured** from **who posts**. Because you cannot follow the same users across platforms (as you honestly note in your limitations), your platform effect is technically an upper bound. Your Step 4 (multilevel modeling holding topic mix constant) needs to be incredibly robust to convince them that you've controlled for demographics as much as observationally possible.
2. **The Validity of the "Anchor Items":** For test equating (Step 5), you plan to use shared policy events as anchor items. Reviewers will ask: *Is a policy announcement genuinely a stable anchor?* On Twitter, an announcement might trigger immediate emotional reactions, while on Reddit, it might spark a nuanced policy debate three days later. You must defend why these events can serve as invariant anchors across fundamentally different communication styles.
3. **The Size of the Yardstick:** You mention using your hand-labelled sample (~200 posts) as your independent validation yardstick. For a dataset of 279k posts, 200 is quite small. Expanding this human-verified sample across all four platforms and ensuring it is evenly distributed will be vital to making your generalizability-theory decomposition reliable.



The Verdict

Your conceptual framing is rock solid. You are introducing a **reusable diagnostic framework** to computational social science. If you prove this method works, any future researcher trying to combine social media platforms into a singular "public opinion index" will have to run your invariance check first. That is a textbook definition of a meaningful academic contribution.

Given that your BERT classifier and data pipeline are already complete, how do you plan to structure the multilevel modeling in Step 4 to ensure the platform effect isn't being quietly driven by unobserved user demographics?

Prompt 2

first, this is not my paper. a colleague asked me if their student's proposal might be fit for ICWSM. some of the papers cited by the study have been published at ICWSM, but I haven't seen such papers recently (I remember O'Connor's paper from 2010. is ICWSM still the best place for this paper? I think the 200K posts across four domains is small, especially if there are no identified events yet.

That context changes the evaluation significantly.

To answer directly: **ICWSM is still the absolute premier venue for this type of computational social science work, but this specific paper proposal, in its current state, will face a very steep uphill battle.** Your instincts regarding the sample size and the structural timeline are entirely correct. The field has moved drastically since O'Connor's seminal 2010 paper (*"From Tweets to Polls"*).

An analysis of why ICWSM is still the right venue, where this proposal falls short, and how the student can fix it to match modern expectations reveals the following key insights:

1. Is ICWSM still the right place?

Yes. ICWSM remains a core hub for blending social science theories with computational methods. If you look at the recent ICWSM conference programs, they explicitly call for papers addressing *"Subjectivity in textual data; sentiment analysis"* and *"Cross-platform studies."* Papers exploring "algorithmic curation effects" or cross-platform differences (e.g., comparing Twitter vs. Reddit) are published there every year.

However, the bar for what constitutes a "contribution" has changed. In 2010, simply proving that social media sentiment correlated with consumer confidence or polling data was groundbreaking. Today, ICWSM reviewers consider basic sentiment correlation over time to be a solved problem (or a baseline). They expect deep methodological rigor, causal identification, or massive scale.

2. The Size Dilemma: 279K Posts across 4 Platforms is Small

You are completely right to call out the dataset size.

- Dividing 279,000 posts across 4 platforms (Twitter, Reddit, YouTube, Quora) averages out to only **~70,000 posts per platform**.
- If those are tracked over a standard policy timeline (say, a year), that breaks down to a incredibly sparse daily volume per platform.
- For a modern ICWSM paper using standard NLP pipelines, reviewers routinely expect datasets in the millions or tens of millions of posts to ensure that the volume is resilient against the extreme noise inherent to social media text.

3. The Major Flaw: "No Identified Events Yet"

This is the paper's critical vulnerability. The outline states that the student plans to use "shared policy events as anchor items" to test for measurement invariance (Step 5).

If they have not pre-identified those events, **the dataset might not actually contain them in sufficient volumes.** * **The "Zero-Cell" Problem:** Economic policy discussions on Quora or YouTube behave entirely differently than on Twitter. A major fiscal announcement might generate 20,000 tweets in an hour, but result in only 5 structural threads on Reddit and zero relevant YouTube videos.

- Without knowing exactly what these "anchor events" are, it is impossible to know if the 279K corpus has enough overlapping data points across *all four* platforms to mathematically calculate Generalizability Theory variance or Differential Item Functioning (DIF).

How the Student Can Salvage This for ICWSM

If the student wants to target ICWSM, they should pivot away from treating this as a massive empirical discovery paper and lean into it as a **Methodological or Dataset Paper**. ICWSM has specific submission tracks for these:

1. **Focus heavily on the Measurement Invariance Framework:** The actual novelty isn't the economic finding; it's the application of psychometric "survey mode" constraints to machine-labelled text. The student should frame the paper entirely around *proving* how dangerous it is for other researchers to aggregate cross-platform text without accounting for platform bias.
2. **Anchor around specific, high-volume dates:** Instead of broad data collection, they should restrict their corpus to highly specific, explosive economic shocks (e.g., the UK "Mini-Budget" crisis, major inflation rate announcements, or specific Autumn Statements). They can then hyper-collect data strictly around those windows to artificially increase the density of their "anchor items."
3. **Consider the Poster Track:** If the sample size cannot be expanded, this work is an excellent candidate for the **ICWSM Poster Paper track**, which explicitly welcomes

smaller-scale studies, preliminary results, and novel frameworks that are still proving out their data pipelines

Prompt 3

I found the master thesis that the student wants to expand, does the data collection chapter explain if the data are comparable across platforms?

Yes, the data collection chapter explicitly addresses this, but it flags it as a limitation rather than providing a deep structural solution.

In **Section 5.5 (Limitations of Data Collection)** under the heading "**Cross-Platform Integration Challenges**," the dissertation notes the following regarding cross-platform data comparability:

1. **Inherent Scale and Style Differences:** The text explicitly acknowledges that there are *"differences in sentiment scale and engagement style"* across Twitter, Reddit, YouTube, and Quora.
2. **The "Band-Aid" Solution Used:** To handle these differences within the master's dissertation itself, the student states that **"proportional metrics [were] used for fair comparisons."**

What this means for the paper expansion:

This confirms why the student's new research proposal pivots toward psychometrics (Measurement Invariance, Generalizability Theory, and Differential Item Functioning). In the master's thesis, they simply used basic mathematical proportions to compare platforms on a surface level. They recognized that a tweet's engagement style is fundamentally different from a lengthy Quora answer, but they didn't have the methodological framework to statistically adjust for it yet.

The fact that the data collection chapter leaves this gap wide open is exactly what justifies expanding this into a full research paper—it proves that standard cross-platform NLP studies usually gloss over platform bias using basic proportions, which is the exact problem their new proposal targets.

Prompt 4

but, is their existing dataset appropriate for the claims they are making? how many messages per post do they have? don't they need a few events to showcase their approach?

The existing dataset presents significant challenges for the claims in the new proposal, and your concerns about the data breakdown and the lack of pre-identified events are entirely correct.

1. How many messages per post do they have?

Looking at the data ingestion schema and details in the dissertation, the student collected **"posts and comments"** combined, rather than tracking comment-to-post depth explicitly. The total corpus of **279,000 entries** is a mixture of top-level posts, replies, and answers, broken down across the four platforms as follows:

- **Twitter (X):** ~120,000 entries (tweets and replies).
- **Reddit:** ~109,000 entries (posts and comments across targeted UK subreddits).
- **YouTube:** ~25,000 comment entries (replies to policy-related video content).
- **Quora:** ~25,000 entries (questions and answers combined).

Because the student treated each comment or reply as an individual, flat text string for sentiment classification, they do not have a robust metric for "messages per post." This means they cannot evaluate engagement depth or community conversational structure; they are simply aggregating independent sentences over time.

2. Is the existing dataset appropriate for the claims they are making?

No, not for the new proposal's advanced psychometric claims. While this dataset was completely appropriate for a Master's dissertation mapping basic sentiment shifts over time, it is structurally deficient for the proposed expansion for several reasons:

- **Sparsity in Daily Volume:** When a dataset spanning multiple years (2018 to 2025) is broken down across 4 platforms and 4 policy topics (Taxation, Welfare, Inflation, Stimulus), the data becomes incredibly thin on a daily level. For example, 25,000 total posts over 7 years on Quora averages out to just **~10 posts per day across the entire platform** for all economic topics combined. Trying to calculate Generalizability Theory variance or Differential Item Functioning (DIF) on such micro-volumes will yield mathematically unstable and noisy results.
- **The "Hierarchy Bias" Problem:** A tweet is limited to a few characters, a Reddit post is a nested paragraph thread, a Quora response is an essay, and a YouTube comment is often short and reactive. Because they do not have a standard "unit of measurement" (some platforms are heavily post-driven, others comment-driven), they cannot easily treat them as identical parallel test items without substantial statistical pre-processing

that the dataset cannot currently support.

3. Don't they need a few specific events to showcase their approach?

Yes, absolutely. To test for measurement invariance and run test equating (Step 5 of the new proposal), they **must** have highly synchronized, distinct external shocks (e.g., specific Autumn Statements or emergency budget announcements) that reverberate across all four platforms at the exact same time.

In the original dissertation, the student relied on broad, macro-level keyword filters covering long periods. The model did happen to pick up a major aggregate downturn during the **September 2022 UK mini-budget crisis**, which proves that a major event *can* be found in the historical data.

However, to make this an acceptable ICWSM paper, the student cannot just rely on one or two historical accidents. They need a systematic set of 5 to 10 distinct "anchor events." They will need to rewrite their data collection strategy to ditch the sparse multi-year timeline and instead hyper-collect data within strict 2-week windows surrounding specific major UK fiscal events. This is the only way to ensure they get the dense, overlapping text volume required to properly showcase and validate their psychometric framework.

Prompt 5

earlier you wrote:

Focus heavily on the Measurement Invariance Framework: The actual novelty isn't the economic finding; it's the application of psychometric "survey mode" constraints to machine-labelled text. The student should frame the paper entirely around proving how dangerous it is for other researchers to aggregate cross-platform text without accounting for platform bias.

Is there a way to do this with the existing dataset, or one really needs to create a new dataset?

The short answer is: **They do not need to create a whole new dataset from scratch, but they cannot simply use the existing dataset "as-is" either.** They can leverage their existing corpus if they retroactively apply a **"Filter & Sub-sample"** methodology.

To move the paper from a descriptive Master's thesis into a rigorous, publishable measurement framework using the data they already have, the student needs to transition from analyzing the *entire timeline* to conducting a series of **Targeted Case Studies**.

How to use the existing dataset to build the argument:

1. Re-frame the Data Around the "September 2022 UK Mini-Budget Crisis"

The dissertation notes that their dataset successfully captured a massive, distinct aggregate peak in negative sentiment across platforms during the late-2022 mini-budget. This is their goldmine. Instead of trying to prove measurement invariance over a thin, noisy multi-year average, they should isolate **just the data from September 15 to October 15, 2022**. Within this strict window, the "economic shock" was a constant external event felt by the entire UK public. By filtering down to just this high-density window, they can evaluate whether Twitter, Reddit, YouTube, and Quora tracked the explosion of negativity identically, or if specific platforms suffered from "Differential Item Functioning" (e.g., did Quora stay unusually neutral while Twitter exploded, and was that due to platform bias?).

2. Introduce an "Anchor Event" Multi-Level Model

To address the concern about not having pre-identified events, the student can use public historical records (like the Bank of England rate announcement dates or official Autumn Statement days between 2018 and 2025) and map those dates directly onto their database. They can pull out 5 or 6 of these specific 48-hour windows. They can then use a Multi-level Regression model where:

- **Level 1:** Individual posts (sentiment score).
- **Level 2:** The specific Anchor Event (e.g., "Budget Day 2024").
- **Fixed Effects:** Platform type (Twitter vs. Reddit vs. YouTube vs. Quora).

If the "Platform Type" remains a statistically significant predictor of sentiment *even when holding the specific event and policy topic constant*, they have successfully used their existing dataset to prove their core thesis: **platforms introduce an algorithmic or stylistic measurement bias that distorts the true public signal.**

3. Quantify the "Danger" via Simulating Aggregation Errors

To prove *"how dangerous it is for other researchers to aggregate cross-platform text without accounting for platform bias,"* they can run a simulation on their current data:

- **The Naive Approach:** Merge all 279,000 posts together into a single "UK Economic Sentiment Index". Because Twitter has 120,000 posts and Quora only has 25,000, the final index will completely drown out Quora's voice.
- **The Measurement Approach:** Apply a basic psychometric adjustment (like weight-adjusting for platform volume or standardizing scores based on each platform's historical baseline).
- **The Proof:** Show how the Naive Index miscalculates the timing or severity of public backlash compared to the adjusted framework. For example, show a graph where a policy looks "highly negative" overall, but it's actually just a tiny, hyper-vocal group on Twitter skewing the aggregate pool while three other platforms are calm.

The Catch: They will likely need a *small* patch of new data

While the historical text is there, their human-validated "yardstick" is too weak. The outline states they only have **200 hand-labelled posts** ($\kappa=0.78$).

To make this a publication-grade ICWSM paper, they don't need a new automated dataset, but they **must expand their human annotation pool**. They should take a random sample of ~1,000 to 1,500 posts *strictly from those newly isolated anchor-event windows* (evenly split: ~300-400 per platform) and hand-label them. This human gold-standard sample is what they will use to mathematically prove that the BERT or VADER classifiers perform differently across platforms due to text length and styling—which acts as the ultimate proof of measurement non-invariance.